

- 2 (a)** A satellite of mass 500 kg moves in a circular orbit at constant speed around the Earth, at a radius r of 9.0×10^6 m, with a period T . The satellite is then moved such that it proceeds to orbit around the Earth at a new radius of 1.0×10^7 m.

The mass of the Earth is 5.97×10^{24} kg and may be assumed to be a point mass at the centre of the Earth.

- (i)** Determine T .

$T = \dots\dots\dots$ s [2]

- (ii)** Determine the work done required in moving the satellite to its new orbit.

work done = $\dots\dots\dots$ J [2]

- (b) (i)** Explain why a geostationary satellite must orbit in the equatorial plane.

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..... [3]

- (ii)** Explain why geostationary satellites are often used for telecommunication.

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..... [1]

3 (a) (i) Explain what is meant by the *internal energy* of a system